

Saksham Adhikari

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EDUCATION

Texas State University

B.S. in Computer Information Systems, AI Concentration, GPA: 3.83/4.0

San Marcos, TX

May 2027

- **Coursework:** Data Structures & Algorithms, Database Systems, Machine Learning, Data Mining in R
- **Achievements:** **6x Hackathon Winner**; Most Commercializable Hack – AWS Hackathon; Winner – AITX NVIDIA Hackathon; **Gold Medal**, Nepal Economics Olympiad; **\$48k** in scholarships; Google TPU Student Researcher Grant

PROFESSIONAL EXPERIENCE

Machine Learning Engineer Intern

AskSLM

May 2026 – Aug 2026

Austin, TX

- Quantized custom-trained **LLaVA** vision models for **edge devices**, monitoring weight clustering and achieving a **3x** increase in image-to-text sequence recognition throughput
- Enhanced internal Retrieval-Augmented Generation (**RAG**) pipeline by implementing **Reciprocal Rank Fusion**, boosting retrieval precision and achieving a **3%** improvement in **Mean Reciprocal Rank (MRR)**
- Engineered real-time video analytics pipelines via **DeepStream SDK**, optimizing frame-rate throughput and end-to-end inference latency using **Harbor** for systematic evaluation

Software Engineer Intern

AskSLM

Jan 2026 – May 2026

Austin, TX

- Fine-tuned a **LLaMA 8B** model with **3,000** curated QA pairs from internal documentation, integrating a **RAG** architecture for factual grounding and securely deploying via **mTLS** encryption
- Migrated and optimized the internal **vLLM** serving stack to fit **NVIDIA DGX Spark VRAM constraints**, redesigning inference configuration to achieve a **2.4x throughput increase** while operating within hardware memory limits
- Integrated the **RAG** agent with **Slack** for daily utilization on NVIDIA **DGX Spark** devices, reducing developer documentation search time by **30%**

Research Assistant

Translational Health Research Center

Jul 2025 – Nov 2025

San Marcos, TX

- Co-authored **2** published papers, building a data pipeline that reconciled U.S. Census datasets containing **~1.2M** rows
- Identified **7** underserved geographic clusters and modeled disease spread across **260,000** patients using multivariate and **Graph Neural Network** models; presented findings to non-technical partners

Software Development Intern

Breast Cancer Resource Center

May 2025 – Jul 2025

Austin, TX

- Built a **React** patient-facing **chat agent** backed by a **RAG** pipeline, reducing caregiver workload by **20%** while improving response grounding and hallucination reduction
- Improved CSV processing speeds from **1900ms** to **1200ms** using asynchronous **coroutines**, enabling faster billing calculations while adhering to internal compliance requirements

Student Software Developer

Texas State Center for Analytics and Data Science

Dec 2024 – May 2025

San Marcos, TX

- Built a hybrid **RAG** pipeline pairing **FAISS** dense vector search with cosine-similarity ranking for AI-driven grant proposal optimization, reducing faculty evaluation time by **80%**
- Managed the TXST **grant pipeline** on a TXST **Red Hat Enterprise Linux** test server via a custom **GitLab CI/CD** pipeline with linting and dependency checks gating every merge, cutting runtime to under **5 minutes** (**21%** reduction)

PROJECTS

Giftmaxxing | Django, Swift, Terraform

2026

- Won **\$6k+** and the **Most Commercializable Hack** at the **AWS v0 Hackathon**, then iterated the application toward release
- Engineered a **shopping agent** and trained a **two-tower ranking model** with real-time item retrieval

TorchLeet | PyTorch, Deep Neural Networks, Tensor Math

2026

- Core contributor to TorchLeet (**LeetCode for PyTorch, 2.5k stars**): **6** PRs merged across frontend fixes and technical answers

LEADERSHIP

ACM AI @ TXST – Vice President

December 2024 – August 2025

- Grew the community of AI developers on campus to **100+** people and organized **10+** technical sessions on MCPs, CLI tools, and OpenClaw

TECHNICAL SKILLS

Languages: Python, Go, Swift, R, Bash, JavaScript, SQL

AI/ML: PyTorch, TensorFlow, Scikit-learn, LLMs, RAG, Fine-tuning (LoRA, QLoRA), Quantization, Distillation, RRF, FAISS

Systems & Infra: Distributed Systems, Concurrency, GCP, AWS, Docker, Kubernetes, CI/CD, Linux, vLLM, DeepStream